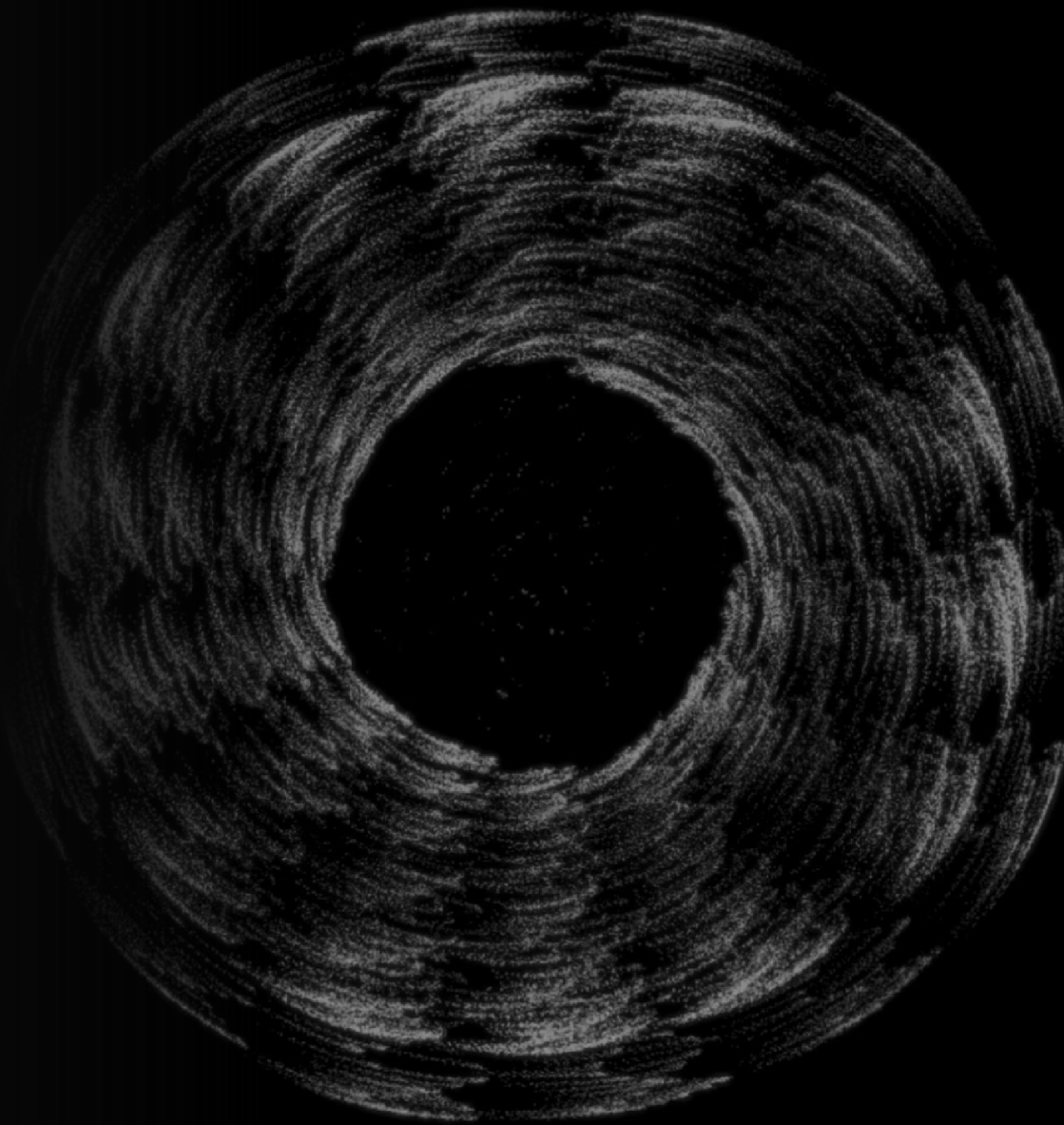
PORTFOLIO · SELECTED WORKS 2020–2026

Chen
Yi-Yen

陳　羿　言

Interactive installation ·
Real-time visuals · Sound ·
Physical computing

互動裝置・即時影像・聲音・軟硬體整合



MFA TECHNOLOGY ART, NTHU · 2026
CYY.YI.YEN@GMAIL.COM · GITHUB.COM/CYIYEN

Works 2020–2026

My MFA research comprised three works examining the physical conditions required for a message to be transmitted. The final work, Beacon Silence, transmits seven channels of sound through custom Li-Fi circuits and was exhibited at VA Hub, Taipei, in July 2026. Production work includes stage video for a theatre performance, two fulldome pieces and the opening transition for a campus presentation at the Future Vision Lab showcase, image generation and installation for *Solar* in LuxuryLogico's solo exhibition at MoCA Taipei, and partial development and testing of the directional-speaker tracking system for *Echo Chamber* at the C-LAB Sound Art Festival. Primary tools: TouchDesigner, Blender and Premiere Pro.

碩士階段以三件作品探討訊息傳遞所需的物理條件。其中《Beacon Silence》以自製 Li-Fi 電路傳輸七聲道聲音，2026 年 7 月於台北 VA Hub 展出。另參與團隊作品及其他創作者作品的製作，包括劇場演出舞台影像、未來視覺實驗室穹頂成果呈現中兩段穹頂作品的內容製作與開場轉場、豪華朗機工台北當代藝術館個展中《日光域》的影像生成與布展，以及 C-LAB 聲響藝術節作品《回聲室》中指向性喇叭追蹤系統的部分開發與測試。常用工具為 TouchDesigner、Blender 與 Premiere Pro。

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01 **Beacon Silence** 2026
INTERACTIVE SOUND INSTALLATION · LI-FI

02 **Paper Shell Echo 紙殼迴聲** 2025
TACTILE SOUND / VR INSTALLATION

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KINETIC OPTICAL INSTALLATION

04 **Fu Pi 負皮** 2024
SENSOR-DRIVEN PERFORMANCE · NATIONAL TAICHUNG THEATER LAB X

05 **Soundscape City 音景城市** 2024
AUDIO-REACTIVE GENERATIVE VIDEO · SHORT STUDY

06 **Compound Eye of Technology 科技複眼** 2023
VR ENVIRONMENT · BLENDER → UNITY

07 **Luminous Membrane** 2023
BIO-ART SCULPTURE · 3D PRINT, SOFT MEMBRANE, LIGHT

08 **The Infinite Depth 無盡深邃的那端** 2022
EXPERIMENTAL THEATRE · MEDIA × DANCE

S **Real-time studies 2026 即時影像研究** 2026
BOTANIC · FULLDOME

P **Production work 協作專案** 2025
COSMIC SKETCHES · ECHO CHAMBER · DOME VISUALS

E **Earlier work 早期作品** 2020–22
NAKED HOUSE · QUANTUM TRANSMISSION PHONE BOOTH · WHO GOD YOU

— **Curriculum vitae**
EDUCATION · EXHIBITIONS · PAPERS

SKILLS

REAL-TIME & GENERATIVE

TouchDesigner (POP, GLSL), Unity, StreamDiffusion, fulldome pipelines, multi-screen playback

3D & PRODUCTION

Blender, Maya, 360°/fisheye rendering, stage and venue modelling, Premiere, Ableton Live

PHYSICAL COMPUTING

Arduino, custom PCB, sensors, lasers & Li-Fi, DMX, motorised mounts, camera tracking

PRODUCTION

Stage video for theatre, fulldome content, exhibition imagery and installation



01 · 2026 · INTERACTIVE SOUND INSTALLATION · LI-FI

Beacon Silence

Seven laser beams each transmit one part of a composition; when a beam is blocked, its part falls silent.

七道雷射各自傳輸一個聲部；光束被遮斷時，對應聲部隨之靜音。

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Seven green laser beams cross a darkened corridor. Each beam carries one part of the same composition to a solar panel on the opposite wall, where it is converted back into sound. The piece is complete only when all seven beams arrive.

The work requires no deliberate interaction. On entering the corridor, a visitor's body interrupts one or more beams, and the corresponding parts drop out. There is therefore no ideal listening position; what each visitor hears depends on where they stand.

The more people present, the quieter the corridor becomes. When all seven beams are blocked, the space is full and the sound is gone. The work defines this state as fading rather than failure: the system continues to run, and what remains audible is silence.

七道綠色雷射橫越暗室走道，每道光束各自承載同一首樂曲的一個聲部，由對面牆上的太陽能板接收並還原為聲音。七道光束全部抵達時，樂曲才得以完整呈現。

CONCEPT	HARDWARE	SOFTWARE / SOUND	SHOWN
Practice-based MFA thesis on the materiality of message transmission	Custom PCBs, laser modules, solar receivers, motorised rails, Arduino	7-channel composition, audio-to-light modulation, web 3D simulator	VA Hub, Taipei, July 2026 · project website with interactive simulator

MEDIUM LI-FI (VISIBLE LIGHT COMMUNICATION), LASER, SOLAR CELL, CUSTOM PCB, ARDUINO

SETUP 7 LASER BEAMS · 7 AUDIO CHANNELS · 6.5 × 2 M CORRIDOR

SOUND 7-PART ABSTRACT COMPOSITION, 3′22″

ROLE CONCEPT, ELECTRONICS, COMPOSITION, SPATIAL DESIGN (SOLO WORK)

SHOWN Unseen Circuits, VA HUB, TAIPEI, 1–3 JULY 2026 · MFA THESIS WORK, NTHU



A visitor crossing the corridor intercepts successive beams, altering the mix with each step.



Standing, walking or crouching: any position in the space interrupts a beam.



Transmitter: 12 V green laser module modulated by the audio signal.



02 · 2025 · TACTILE SOUND / VR INSTALLATION

Paper Shell Echo

紙殼迴聲

A cardboard shell replaces the loudspeaker as the sounding body, paired with fulldome imagery in a VR headset.

以紙殼振動取代喇叭發聲，搭配頭戴式顯示器中的穹頂影像。

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以紙殼振動取代喇叭發聲，搭配頭戴式顯示器中的穹頂影像。

Fulldome performances surround the audience with sound, yet the path the sound travels, from speaker to dome surface to seat, usually goes unnoticed. Paper Shell Echo reduces the dome to a cardboard shell; the visitor lies down with their head inside it.

The installation contains no loudspeaker. Linear resonant actuators embedded in the shell convert audio into vibration, so the cardboard itself becomes the sounding surface. Sound reaches the listener partly through the air and partly through the back of the head in contact with the shell.

The headset plays 360° content: two fulldome pieces produced for the 2024 Future Vision Lab showcase, played back in a virtual dome simulator.

穹頂展演以聲音包覆觀眾的身體，但聲音從喇叭、穹幕到座位的傳遞路徑通常不被察覺。《紙殼迴聲》將穹頂縮小為紙板殼體，觀眾躺臥後將頭部置於殼內。

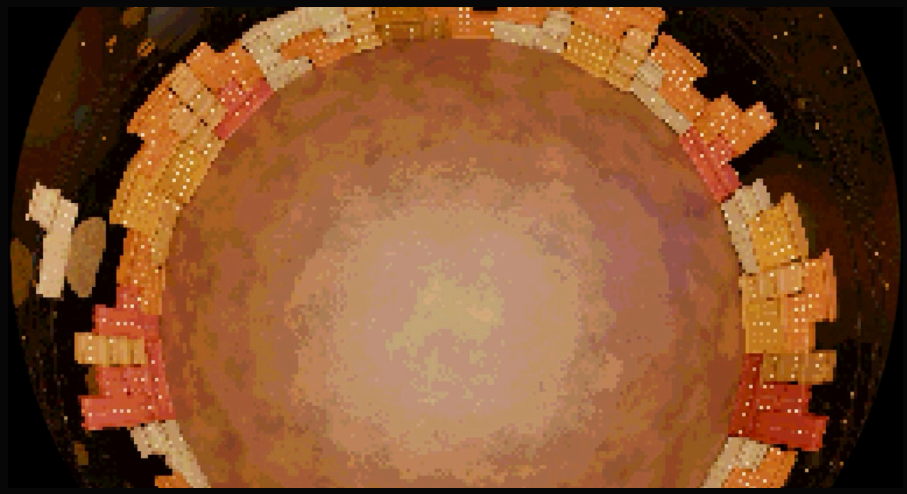
CONCEPT	BUILD	CONTENT	RECOGNITION
Sound conduction as a material interface	Cardboard structure, Taptic Engine LRA array, Bluetooth audio receiver	Two FVL 2024 fulldome pieces, played in a virtual dome simulator	CGW 2025 Honorable Mention (artwork track)

MEDIUM CARDBOARD ENCLOSURE, TAPTIC ENGINE LINEAR ACTUATORS, BLUETOOTH AUDIO, VR HEADSET, 360° VIDEO

FORMAT SINGLE-VISITOR, LYING DOWN, ~4 MIN

ROLE CONCEPT, STRUCTURE, ELECTRONICS, 360° CONTENT

SHOWN CGW 2025 (COMPUTER GRAPHICS WORKSHOP), TAIPEI — HONORABLE MENTION



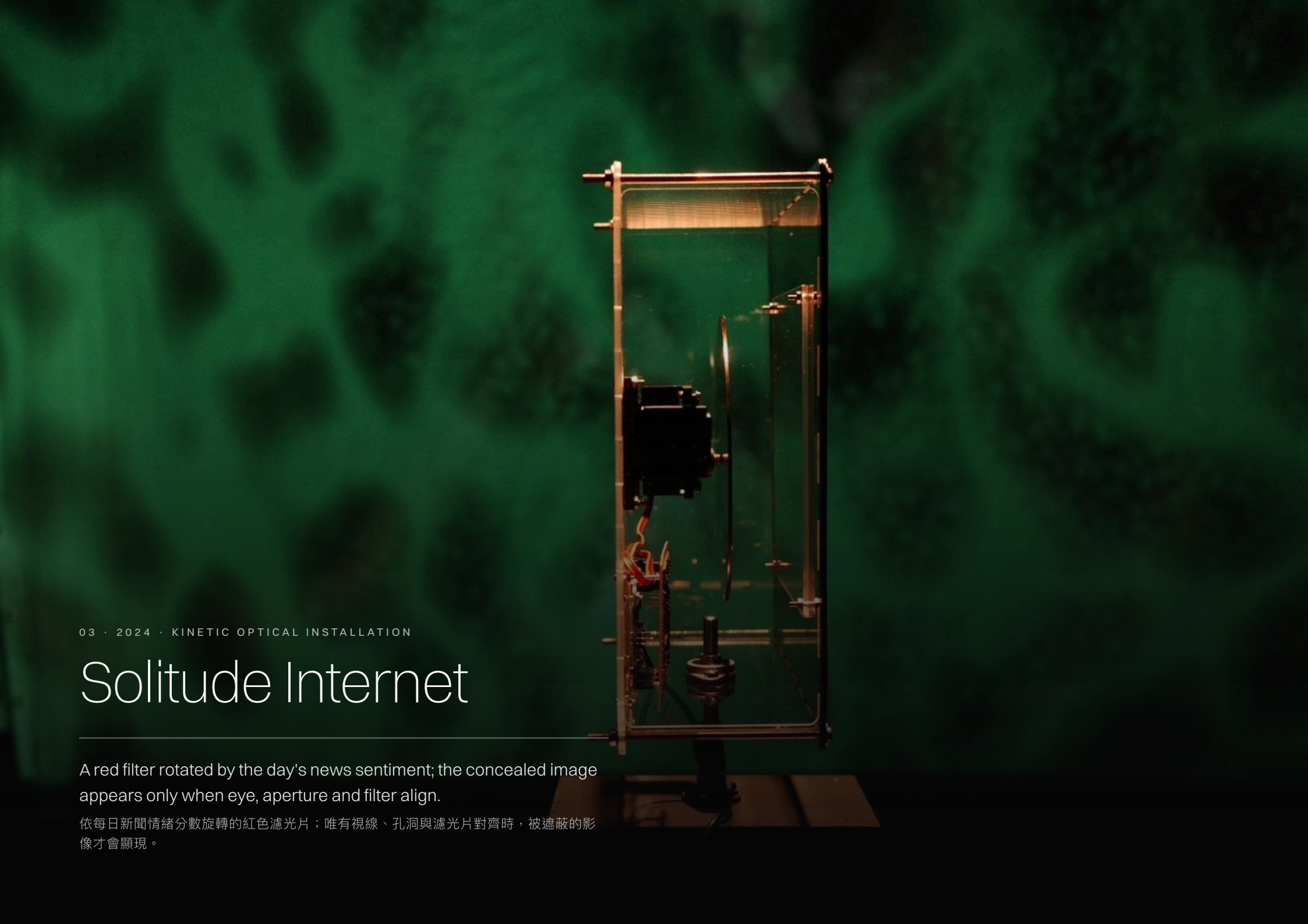
360° content in the headset: the first fulldome piece from the 2024 Future Vision Lab showcase, played in a virtual dome.



The shell: a cardboard enclosure shaped around the head, with actuators concealed in the wall.



Lying inside with the headset on, the visitor's weight presses the head against the sounding surface.



03 · 2024 · KINETIC OPTICAL INSTALLATION

Solitude Internet

A red filter rotated by the day's news sentiment; the concealed image appears only when eye, aperture and filter align.

依每日新聞情緒分數旋轉的紅色濾光片；唯有視線、孔洞與濾光片對齊時，被遮蔽的影像才會顯現。

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A script collects Taiwan's trending news headlines each day, analyses their sentiment and reduces the result to a single score. The score sets the angle of a servo motor, which rotates a red filter inside a small box. The collective mood of the day becomes a slowly turning filter.

Two facing projection screens layer two images: green noise representing information overload, and beneath it a red image of a bare body. To the naked eye only the noise is visible. The red image appears only when the viewer aligns their eye with the aperture in the box at the moment the filter passes in front of it.

Algorithmic filtering normally happens instantaneously in software; this work slows it into an optical process that can be observed, or missed. The Linz version used a wooden box; for Katowice it was rebuilt in clear acrylic to expose the mechanism. The accompanying paper was presented at ISAT 2024.

程式每日擷取台灣熱門新聞標題並進行情緒分析，將結果換算為單一分數，以此控制伺服馬達的角度，轉動盒內的紅色濾光片。社會整體的情緒因此成為一片緩慢轉動的濾鏡。

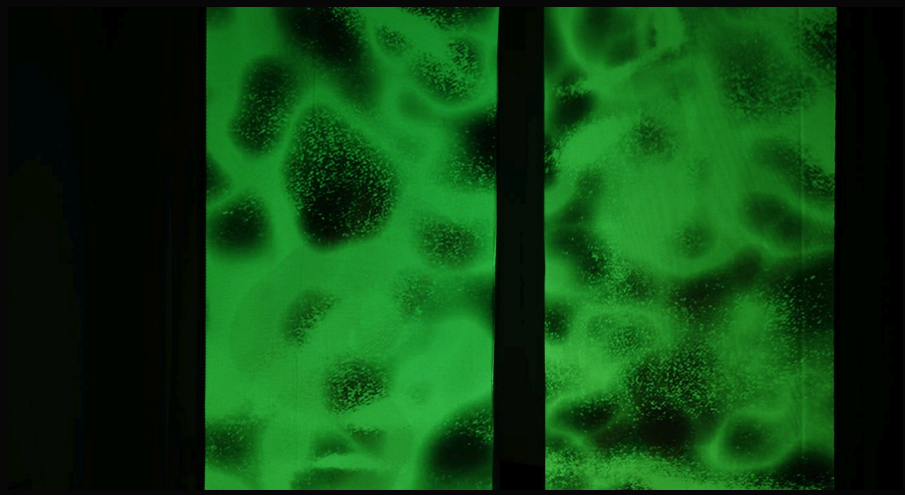
CONCEPT	BUILD	DATA	PAPER
Algorithmic filtering as a material boundary	Arduino Leonardo, MG996R servo, acrylic filter, two-layer projection	Google Trends scraper, translation, VADER sentiment (−1 to +1)	ISAT 2024, Taipei National University of the Arts

MEDIUM BOX WITH ROTATING RED ACRYLIC FILTER, SERVO, ARDUINO, TWO FACING PROJECTION SCREENS, LIVE NEWS SENTIMENT FEED

VERSIONS V1.0 WOODEN BOX 45 × 45 × 10 CM (LINZ) · V2.0 ACRYLIC BOX 27 × 27 × 7.5 CM (KATOWICE)

COLLABORATION CO-CREATED WITH ONE COLLABORATOR

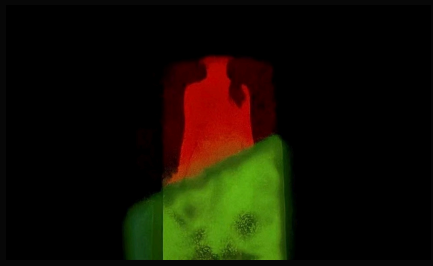
SHOWN 2024 ARS ELECTRONICA FESTIVAL CAMPUS EXHIBITION, POSTCITY LINZ — Anthropogenic Encroachment · NOV 2024 RONDO SZTUKI, KATOWICE — EUROPEAN CITY OF SCIENCE



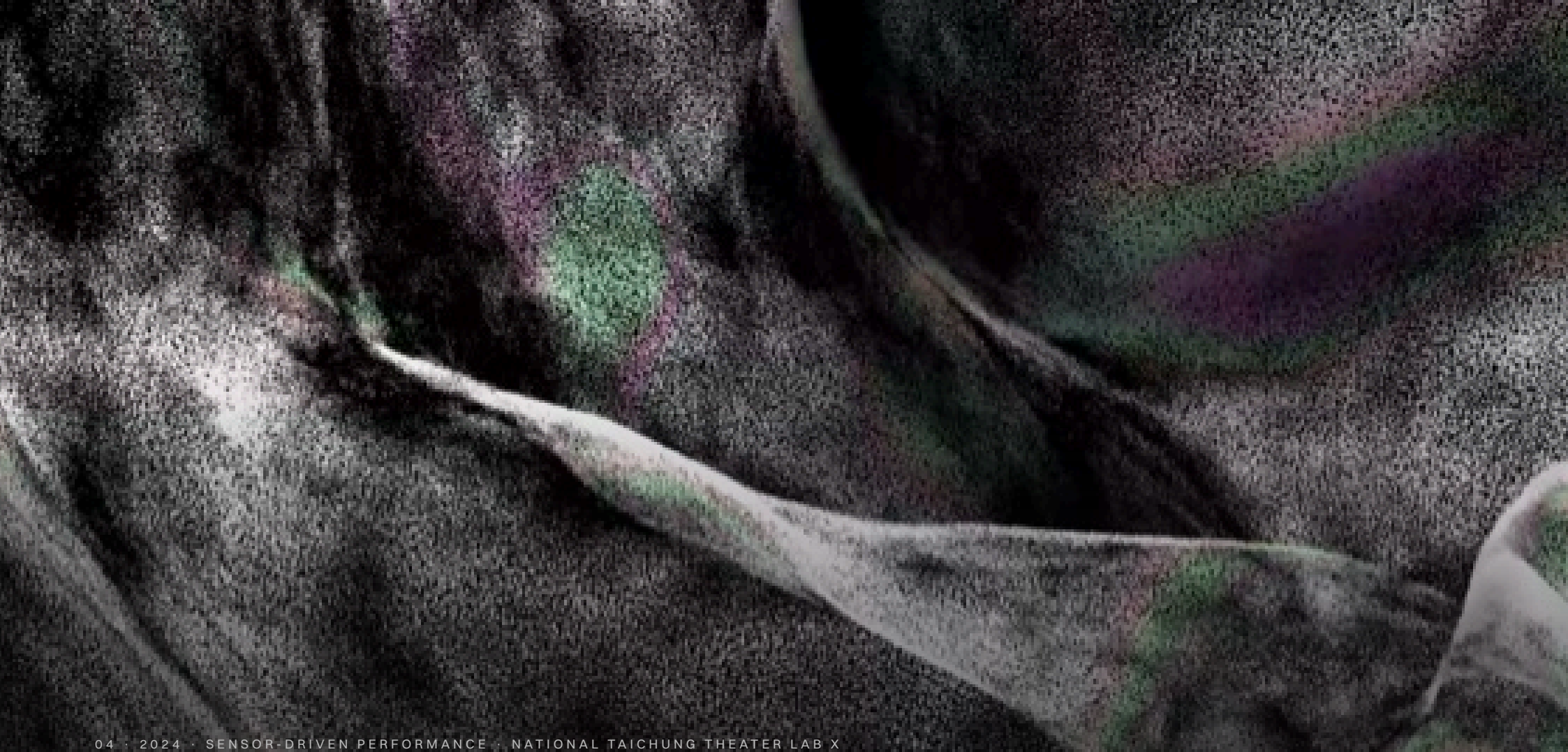
Two facing screens: green noise representing information overload, with the red figure beneath.



The red layer alone, as revealed when filter, aperture and eye align.



View through the aperture: the red figure appears only when filter, aperture and eye align.



04 · 2024 · SENSOR-DRIVEN PERFORMANCE · NATIONAL TAICHUNG THEATER LAB X

Fu Pi

負 皮

A dance performance built around body sensing; responsible for stage video design and production.

以身體感測為主題的舞蹈演出；負責舞台影像的設計與製作。

Fu Pi

A dance performance built around body sensing; responsible for stage video design and production.

以身體感測為主題的舞蹈演出；負責舞台影像的設計與製作。

Fu Pi was selected for the National Taichung Theater's LAB X programme. Drawing on the Liaozhai tale *Painted Skin*, it treats skin as an interface between the physical body and online identity, and asks what remains of a body once it has been read by sensors. A dancer performs behind a translucent curtain while cameras capture the body's outline, heat and skeleton; the data returns to the stage as projected light.

Responsible for the stage video: all cues (0–7) on the dual-screen wall, produced in TouchDesigner and Blender and synchronised with the choreography.

The project ran from June to November 2024, with a public presentation at the theatre on 23 November.

《負皮》為台中國家歌劇院 LAB X 計畫入選作品，取材自《聊齋誌異》〈畫皮〉，將皮膚視為肉身與網路身分之間的介面，探問身體經感測器讀取後所留存的部分。舞者於半透明布幕後演出，攝影機擷取身體的輪廓、溫度與骨架資料，再轉化為投射於舞台上的光影。

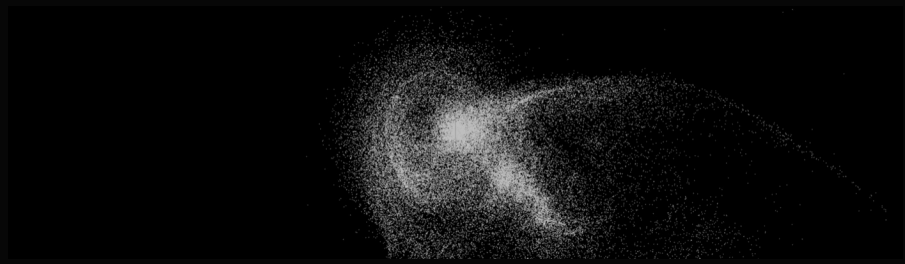
CONTRIBUTION	TOOLS	FORMAT	TEAM
Stage video: all video cues in the performance	TouchDesigner, Blender	Dual screen, 3840 × 1080	Six members; sensing, dramaturgy and performance by other team members

MEDIUM LIVE PERFORMANCE WITH BODY SENSING: DEPTH CAMERA, THERMAL AND SKELETON TRACKING, REAL-TIME PROJECTION, VR

PROJECT TEAM OF SIX; THIS PAGE PRESENTS THE STAGE VIDEO COMPONENT

ROLE STAGE IMAGERY: THE FULL CUE SEQUENCE (CUE 0–7) FOR THE DUAL-SCREEN WALL, 3840 × 1080, BUILT IN TOUCHDESIGNER

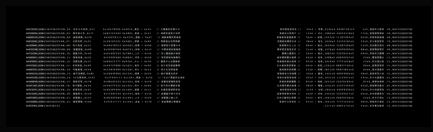
SHOWN 2024 NATIONAL TAICHUNG THEATER LAB X — “YOUNG ARTISTS' STUDIO: BODY SENSING” (SELECTED), WORK PERIOD JUN–NOV 2024, SHOWING 23 NOV 2024



Cue 1



Cue 3: the dancer's scanned body and head rebuilt as particles across both screens.



Cue 3, text layer: sensor readings scrolling as data columns.



05 · 2024 · AUDIO-REACTIVE GENERATIVE VIDEO · SHORT STUDY

Soundscape City

音 景 城 市

A city image generated from ten everyday field recordings.

以十段日常錄音生成的城市影像。

Soundscape City

A city image generated from ten everyday field recordings.

以十段日常錄音生成的城市影像。

Ten everyday recordings, including wind on a sixth floor in Hsinchu, a lighthouse in Kaohsiung and waves at Nanliao Harbour, are edited in Reaper to extract their percussive onsets. Each onset raises a block on a grid, building up a city in TouchDesigner.

The system was also designed to accept spoken keywords: speech recognition extracts the words, and StreamDiffusion regenerates the city image accordingly, layering recordings from different times and places into a single image.

十段日常錄音，包括新竹六樓的風聲、高雄燈塔與南寮漁港的海浪，先於 Reaper 中擷取節奏點。每個節奏點在格狀地圖上生成一個量體，聲音由此在 TouchDesigner 中堆疊為城市。

SOUND	VISUAL	AI	PERIOD
Field recording, Reaper editing, onset extraction	TouchDesigner grid-city generator	Speech recognition → StreamDiffusion (TouchDesigner integration)	Apr–Jun 2024

MEDIUM TEN FIELD RECORDINGS, REAPER, TOUCHDESIGNER, SPEECH RECOGNITION, STREAMDIFFUSION

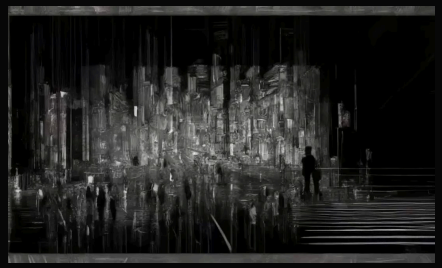
FORMAT SINGLE-CHANNEL GENERATIVE VIDEO

ROLE CONCEPT, SOUND EDITING, TOUCHDESIGNER SYSTEM, AI PIPELINE (SOLO)

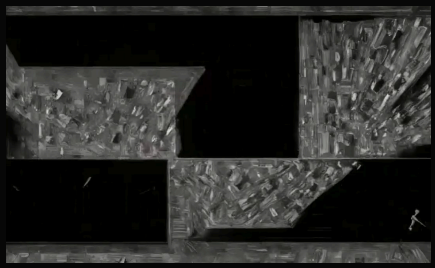
YEAR 2024



Two figures on an illuminated plaza; the moon acts as a sound source.



Figures gather where the recordings are densest.



Blocks rise on the grid with each percussive hit.



06 · 2023 · VR ENVIRONMENT · BLENDER → UNITY

Compound Eye of Technology

科 技 複 眼

A two-scene VR work on personal data online.

以網路個人資料為題的雙場景 VR 作品。

Compound Eye of Technology

A two-scene VR work on personal data online.

以網路個人資料為題的雙場景 VR 作品。

The work treats each online account as an eye that both gathers and disseminates points of view. It comprises two spaces: a circular hall built from symbols of personal and religious attachment, and a mirrored room whose pillars represent stored personal data.

Initially the viewer moves freely and chooses where to look; as the scene changes, their position becomes fixed and their view becomes one of the screens within the room. The masked figures in the hall were modelled in Blender.

作品將每個網路帳號視為一隻眼睛，既收集觀點，也將觀點傳播出去。VR 中包含兩個空間：以個人與信仰符號構成的圓形大廳，以及以柱體代表被儲存之個人資料的鏡面房間。

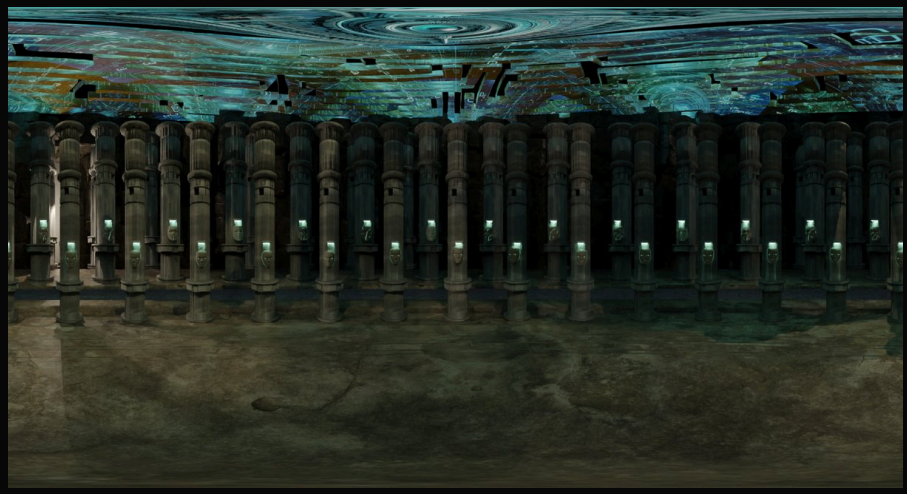
3D	ENGINE	CONCEPT	PERIOD
Blender modelling, baking, 360° renders	Unity, Meta Quest build	Accounts as eyes that collect and pass on views	Sep–Dec 2023

MEDIUM VR (META QUEST), BLENDER MODELLING AND BAKING, UNITY, 360° RENDERS

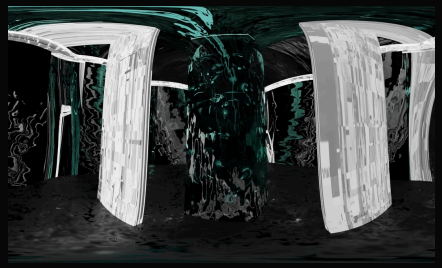
SCENES 1 · A TEMPLE-LIKE ROUND HALL; 2 · A MIRRORED CLOSED ROOM WHOSE PILLARS ARE THE INTERFACE BETWEEN PEOPLE AND THE SYSTEM

ROLE CONCEPT, MODELLING, UNITY BUILD (SOLO)

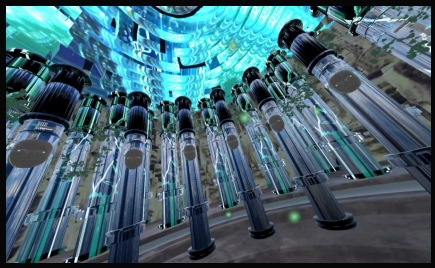
YEAR 2023



Scene 1: the round hall, 360° render.



Scene 2: mirrored room, 360° render.



Quest walkthrough: the figures in the hall.



07 · 2023 · BIO-ART SCULPTURE · 3D PRINT, SOFT MEMBRANE, LIGHT

Luminous Membrane

A 3D-printed sculpture illuminated from within by LED lines.

以 3D 列印成形、由內部 LED 照明的雕塑。

Luminous Membrane

A 3D-printed sculpture illuminated from within by LED lines.

以 3D 列印成形、由內部 LED 照明的雕塑。

The work continues an observation of biological interaction, starting from the encounter between a person and a mosquito within personal space. The anthropologist Edward T. Hall divided the space around a person into four zones: public, social, personal and intimate. Personal space is controlled by the individual, and others need permission to enter. A mosquito, driven by the instinct to survive and reproduce, crosses into it; a person answers with a slap to restore order.

Luminous Membrane turns this invaded private zone into a glowing layer of flesh: in a wall of identical red modules, one holds a soft pink core lit from within, and lines of light extend outward from it to suggest infection and spread after the intrusion.

The core was sculpted in Blender, prepared for 3D printing and lit with Arduino-controlled LED lines. The wall images are renders; the photograph shows the printed piece.

作品延續對生物互動的觀察，從人與蚊子在「個人空間」裡的相遇出發。人類學家愛德華·霍爾（Edward T. Hall）將人周圍的空間分為公共、社交、個人與內部四個層次；個人空間由個體掌控，他人進入需要許可。蚊子出於生存與繁殖的本能闖入這個範圍，人則以拍打維持私密空間的秩序。

3D	FABRICATION	ELECTRONICS	PERIOD
Blender sculpting, STL preparation	3D-printed modules	Arduino, LED lines	Oct–Dec 2023

MEDIUM BLENDER SCULPT, 3D-PRINTED MODULES, TRANSLUCENT MEMBRANE, LEDS, ARDUINO

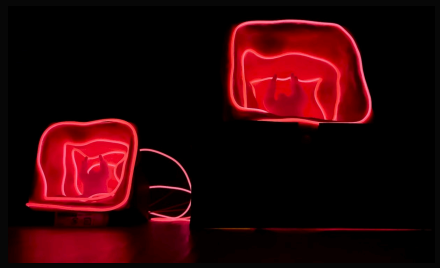
FORMAT WALL OF REPEATED MODULES WITH ONE ILLUMINATED ORGANIC CORE (RENDERS; PROTOTYPE MODULES PRINTED)

ROLE CONCEPT, SCULPTING, PRINT FILES, ELECTRONICS (SOLO)

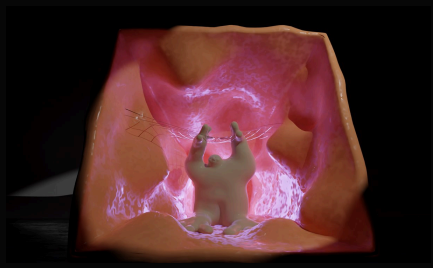
YEAR 2023



The printed piece with LED lines illuminated.



Render of the same lit form against black.



The core, sculpted in Blender.



08 · 2022 · EXPERIMENTAL THEATRE · MEDIA × DANCE

The Infinite Depth

無盡深邃的那端

A dance performance combining a mirror installation with live projection.

結合鏡面裝置與即時投影的舞蹈演出。

The Infinite Depth

A dance performance combining a mirror installation with live projection.

結合鏡面裝置與即時投影的舞蹈演出。

A mirror installation receives the live projection and reflects it in fragments. The dancer interacts with and imitates their own afterimage before being replaced by it, while LCD screens carry a parallel narrative.

The narrative follows a person caught in rumour: how fragments of information on social media assemble an image that does not match them, and the self-doubt that follows.

鏡面裝置承接即時投影，並將其碎裂反射。舞者與自身殘影互動、模仿，最終被殘影取代；液晶螢幕同時呈現另一條敘事線。

SPATIAL DESIGN	VIDEO	TEAM	VENUE
Mirror installation and stage arrangement	Contributed part of the video	Eight members; direction, choreography, video design, sound and stage by other team members	Nao Space, Taipei Fringe Festival 2022

MEDIUM PROJECTION, MIRROR INSTALLATION, VIDEO, DANCE, SOUND AND LIGHT · ~66 M² BLACK BOX

TEAM TEAM OF EIGHT, ELECTRIC WHITE RATS (電氣白老鼠)

ROLE SPATIAL DESIGN (CREDITED ROLE); PART OF THE VIDEO

SHOWN 2022 TAIPEI FRINGE FESTIVAL, NAO SPACE 4F



Live projection, mirror installation and dancer at Nao Space.



Act IV video, shot for the LCD narrative.



Dancer under the lighting design.

Botanic

A generative plant that grows according to the viewer's posture.

依觀者姿勢生長的生成植物。

ENGINE TOUCHDESIGNER 2025, POP PARTICLES, L-SYSTEM

INPUT WEBCAM SILHOUETTE, NUMPY POSE ANALYSIS

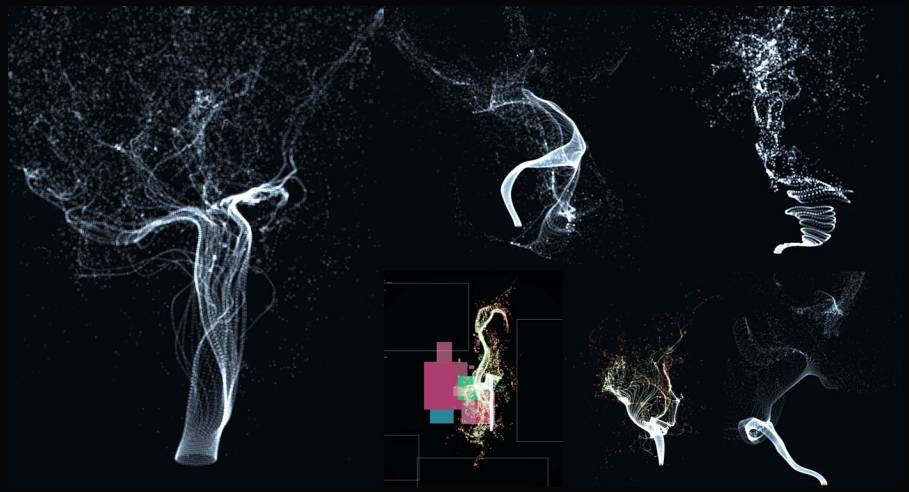
OUTPUT 1280 × 1280 ARTWORK + MONITOR VIEW

STATUS WORKING PROTOTYPE, SEPT 2026

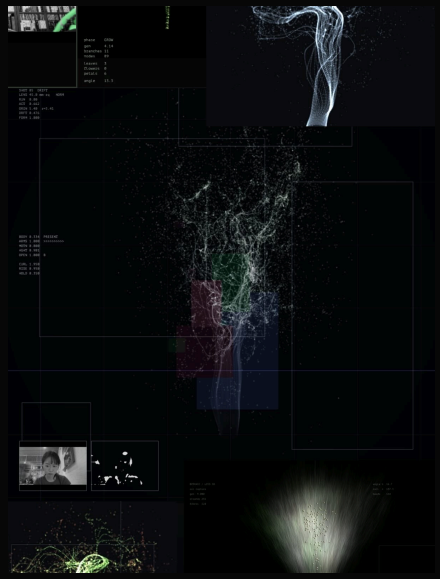
A camera tracks the viewer's posture: bending forward keeps the seedling low and drooping, standing upright makes it grow taller, and opening the arms spreads the branches into bloom. The plant is an L-system of six geometry classes (trunk, branches, nodes, leaves, flowers, petals) driven by three body readings: uprightness, arm spread and crouch.

The silhouette is downsampled to a 32 × 24 grid and measured relative to the body's own bounds, so distance from the camera has no effect. Growth stage, branching angle, gravity and internode length are expressed as functions of these readings; a particle cloud resampled from the skeleton forms the surrounding haze.

攝影機擷取觀者姿勢：彎腰時幼苗低垂，站直時向上生長，張開雙臂時側枝展開並開花。植物以 L-system 建構，由六類幾何（主幹、枝、節、葉、花、花瓣）組成，並由三項身體數值驅動：直立程度、手臂張開程度與蹲低程度。



Point-cloud passes and growth states: the skeleton resampled into particles, colour blocks where the body is read.



Working collage: camera, silhouette, live readings, plant states.



Bend · stand · arms open: the three postures and the plants they grow.

Fulldome 2024–2026

Fulldome content for a 2024 campus presentation, and subsequent generative systems for the dome.

2024 年穹頂校園成果呈現的影像，以及後續的穹頂生成系統。

ENGINE TOUCHDESIGNER, GLSL POP GENERATORS, CUBE-MAP → FISHEYE

FORMAT 1280 × 1280 FISHEYE, 235°, 60 FPS · 4096 × 4096 MASTERS

SCREENED FUTURE VISION LAB 2024, C-LAB DOME THEATRE, 22 DEC 2024 — NTHU × SHIH CHIEN PRESENTATION: TWO PIECES AND THE OPENING TRANSITION

STATUS IN DEVELOPMENT, SEPT 2026

For the NTHU × Shih Chien University student presentation at Future Vision Lab 2024 (C-LAB Dome Theatre, 22 December 2024), a team production, produced two fulldome pieces and the opening transition: a skin-like sphere that swells open, and a black hole that draws particles inward. Both pieces were projected across the full dome above the seated audience. Work in the fulldome format has continued since.

Two systems for fulldome projection followed. The first is a 70-second loop morphing between seven procedural geometries (rings, lattice, tunnel, wave, caustic, depth, spiral) every ten seconds. Because the dome exceeds a standard 180° lens, the image is rendered via a cube map and converted to a 235° fisheye.

The second is a particle study on emotional self-depletion. Particles contract towards a black centre, accelerating as they approach, then rebound outward, each cycle shorter than the last. Layered speeds, depth-dependent particle size and a dark centre give the motion a sense of inward pull.

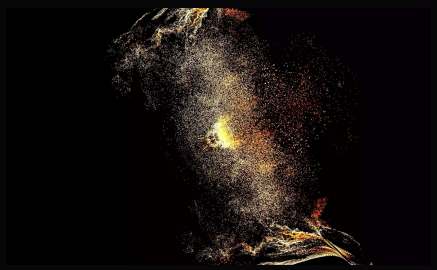
於 2024 年未來視覺實驗室「清華大學藝術學院學士班 / 科藝所 × 實踐大學媒傳系成果呈現」（C-LAB 穹頂劇場，2024 年 12 月 22 日，團隊製作）中，製作兩段穹頂作品與開場轉場：其一為如皮膚般膨脹展開的球體，其二為將粒子收攏的黑洞，皆投映於觀眾上方的整面穹幕。此後持續進行穹頂影像的研究。



The first piece running in the dome, NTHU × Shih Chien presentation, December 2024 — 360° capture from the floor.

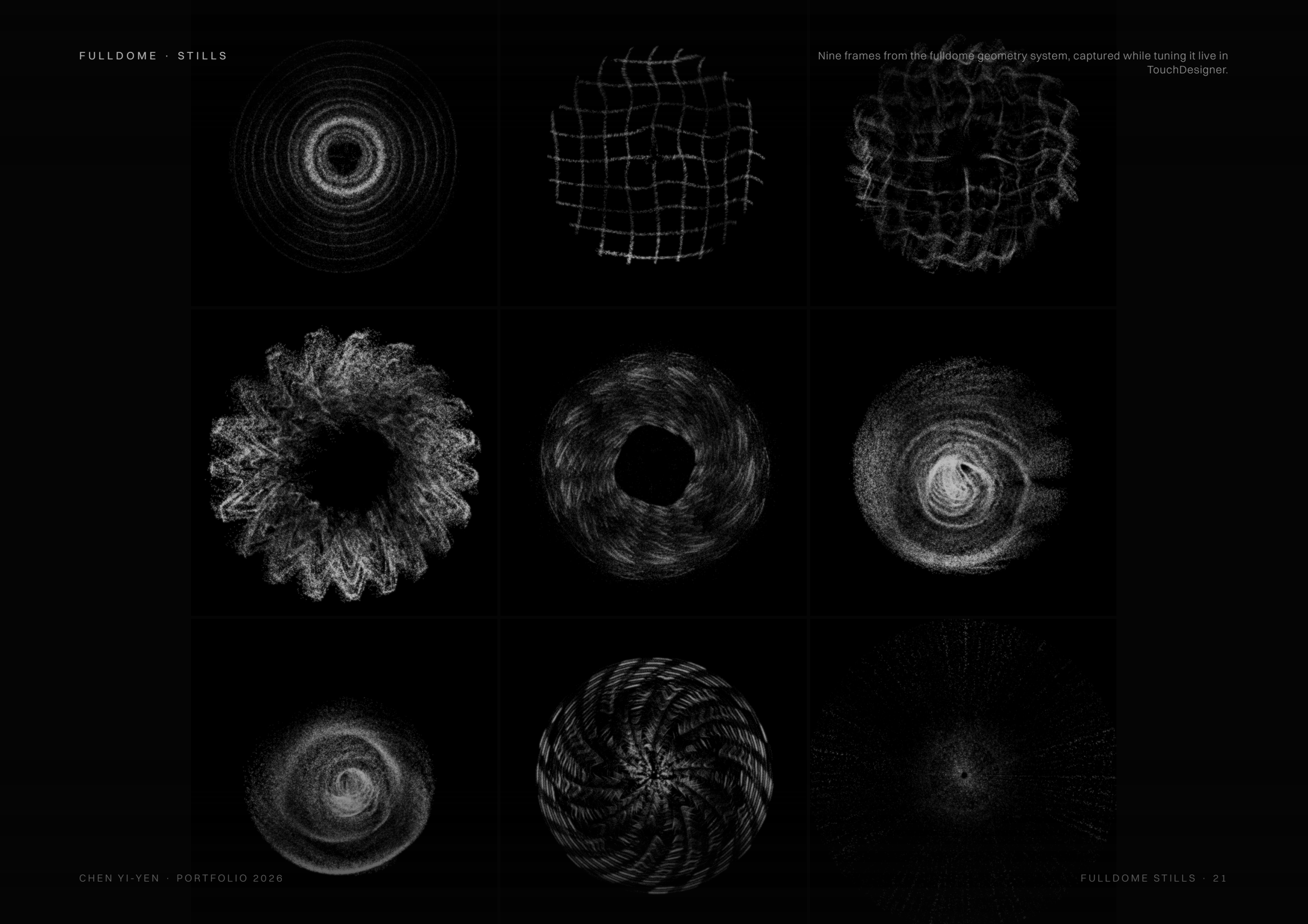


The second piece: particles filling the whole surface.



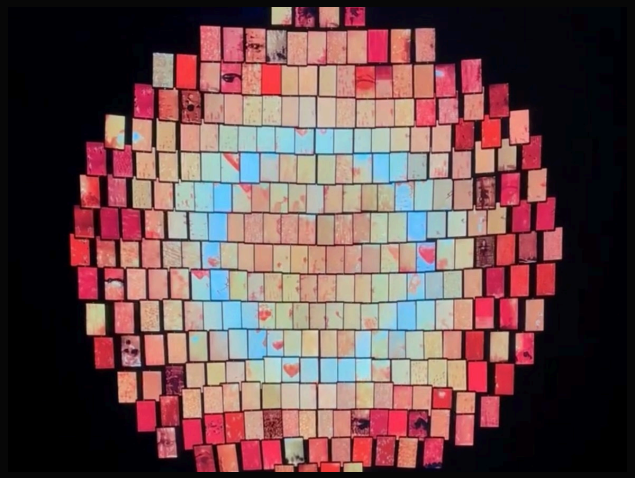
The same sequence as rendered: particles gathered into a black hole.

Nine frames from the fulldome geometry system, captured while tuning it live in TouchDesigner.



Production work

Contributions to productions by other artists.



Cosmic Sketches

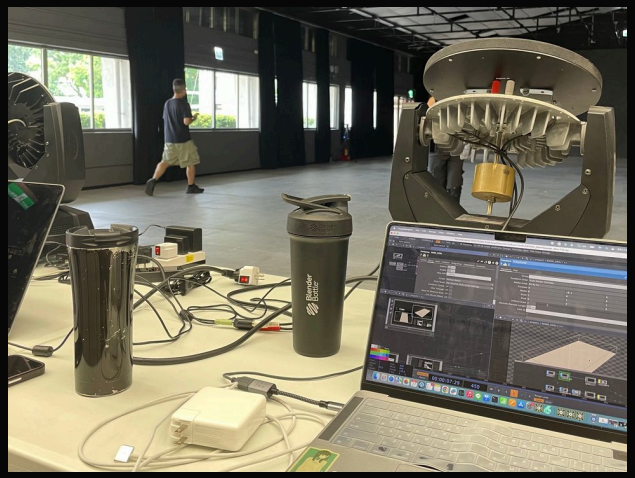
宇宙寫生

LuxuryLogico solo exhibition

Museum of Contemporary Art Taipei · 4 Oct 2025 – 11 Jan 2026

Contribution: Generated image content for *Solar*, and installation on site.

參與內容：《日光域》影像內容生成與現場布展。



Echo Chamber

回聲室

by 吳秉聖, 陳乂 and 王仲堃

2025 C-LAB Sound Art Festival: DIVERSONICS, Taipei · November 2025

Contribution: Took part in developing and testing the tracking system for the directional speakers, built in TouchDesigner.

參與內容：參與指向性喇叭追蹤系統（TouchDesigner）的部分開發與測試。



Dome project visuals

穹頂計畫視覺示意

Group dome project

Taiwan · 2025

Contribution: Visual concept renders for the dome.

參與內容：穹頂畫面視覺示意圖製作。

Earlier work

Undergraduate team projects, Taipei National University of the Arts, 2020–2022.



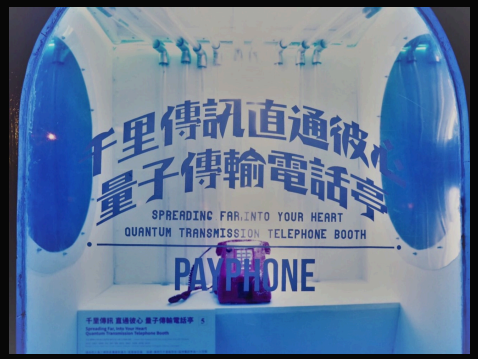
Naked House

感官全裸

2022 · VIDEO INTERACTIVE INSTALLATION

Graduation exhibition, Huashan 1914, Taipei · Blender, Max, Arduino

Max + Arduino integration, Blender character design



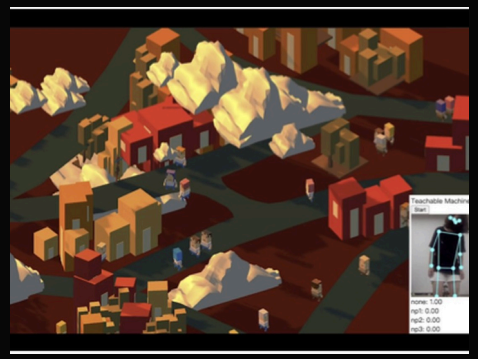
Quantum Transmission Phone Booth

千里傳訊直通彼心 量子傳輸電話亭

2020 · OUTDOOR LIGHT INSTALLATION

C-LAB cross-school workshop · Architecture × New Media × Civil Engineering

Arduino lighting + telephone control, structure fabrication



WHO GOD YOU

基於機器學習之體感骨架辨識遊戲

2021 · MACHINE-LEARNING GAME · RESEARCH DEMO

TAICHI 2021 demo paper · Unity + Teachable Machine

TAICHI 2021, 7th Taiwan CHI conference, Taipei — demo paper

Chen Yi-Yen

陳 羿 言

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Taipei, Taiwan

Tools

TouchDesigner
Stage video for *Fu Pi*; *Soundscape City*; fulldome geometry and particle systems; *Botanic*; directional-speaker tracking system for *Echo Chamber* (partial)

Blender
Modelling for *Compound Eye of Technology*; sculpting and 3D-print files for *Luminous Membrane*; stage video for *Fu Pi*; characters for *Naked House*; 3D models for the portfolio website

Unity
VR build of *Compound Eye of Technology* (Meta Quest); pose-recognition game *WHO GOD YOU*

Arduino / custom PCB
Li-Fi transmitters for *Beacon Silence*; servo-driven filter for *Solitude Internet*; LED lighting for *Luminous Membrane*; voice-reactive lights for *Quantum Transmission Phone Booth*

Max/MSP
Max and Arduino integration for *Naked House*

Python
Pose analysis for *Botanic* (written with AI assistance)

Other
Premiere Pro · Illustrator · Photoshop · After Effects · Maya · Ableton Live

Languages

Mandarin (native) · English (working proficiency) · Japanese (JLPT N2)

EDUCATION

2023–2026 MFA, Institute of Technology Art, National Tsing Hua University
Thesis: *Emergence through Fading: A Practice-Based Study on the Materiality of Message Transmission*

2018–2022 BFA, Department of New Media Art, Taipei National University of the Arts

EXHIBITIONS & PERFORMANCES

2026 *Unseen Circuits*, VA Hub, Taipei (1–3 July) — Beacon Silence

2025 CGW 2025 demo — Paper Shell Echo, Taipei (Honorable Mention)

2024 Ars Electronica Festival Campus Exhibition, POSTCITY Linz — *Anthropogenic Encroachment* (Solitude Internet)

2024 Huashan 1914 Creative Park, Taipei — Solitude Internet 2.0 preview (16–17 Oct)

2024 Rondo Sztuki, Academy of Fine Arts Katowice — European City of Science 2024 (Solitude Internet 2.0)

2024 National Taichung Theater LAB X "Young Artists' Studio: Body Sensing" — *Fu Pi* (selected, team of six; showing 23 Nov)

2024 Future Vision Lab 2024, C-LAB Dome Theatre — NTHU × Shih Chien University student presentation: two fulldome pieces and the opening transition (22 Dec 2024)

2022 *Endless Migration*, Huashan 1914, Taipei — Naked House · Taipei Fringe Festival — *The Infinite Depth* (spatial design, team of eight)

2020 C-LAB *Lightweight* cross-school workshop — Quantum Transmission Phone Booth

EXPERIENCE

2025 *Cosmic Sketches*, LuxuryLogico solo exhibition, Museum of Contemporary Art Taipei (4 Oct 2025 – 11 Jan 2026) — generative imagery for *Solar* and installation build

2025 *Echo Chamber* by 吳秉聖, 陳乂 and 王仲堃, C-LAB Sound Art Festival, Taipei — partial development and testing of the directional-speaker tracking system

2022 Dake Creative Arts, Taipei — arts administration intern (theatre, cross-disciplinary projects)

PAPERS

2025 Chen & Tsai. Paper Shell Echo: a multisensory device combining VR dome immersion with vibration conduction through an enclosing cardboard structure. CGW 2025 — Honorable Mention

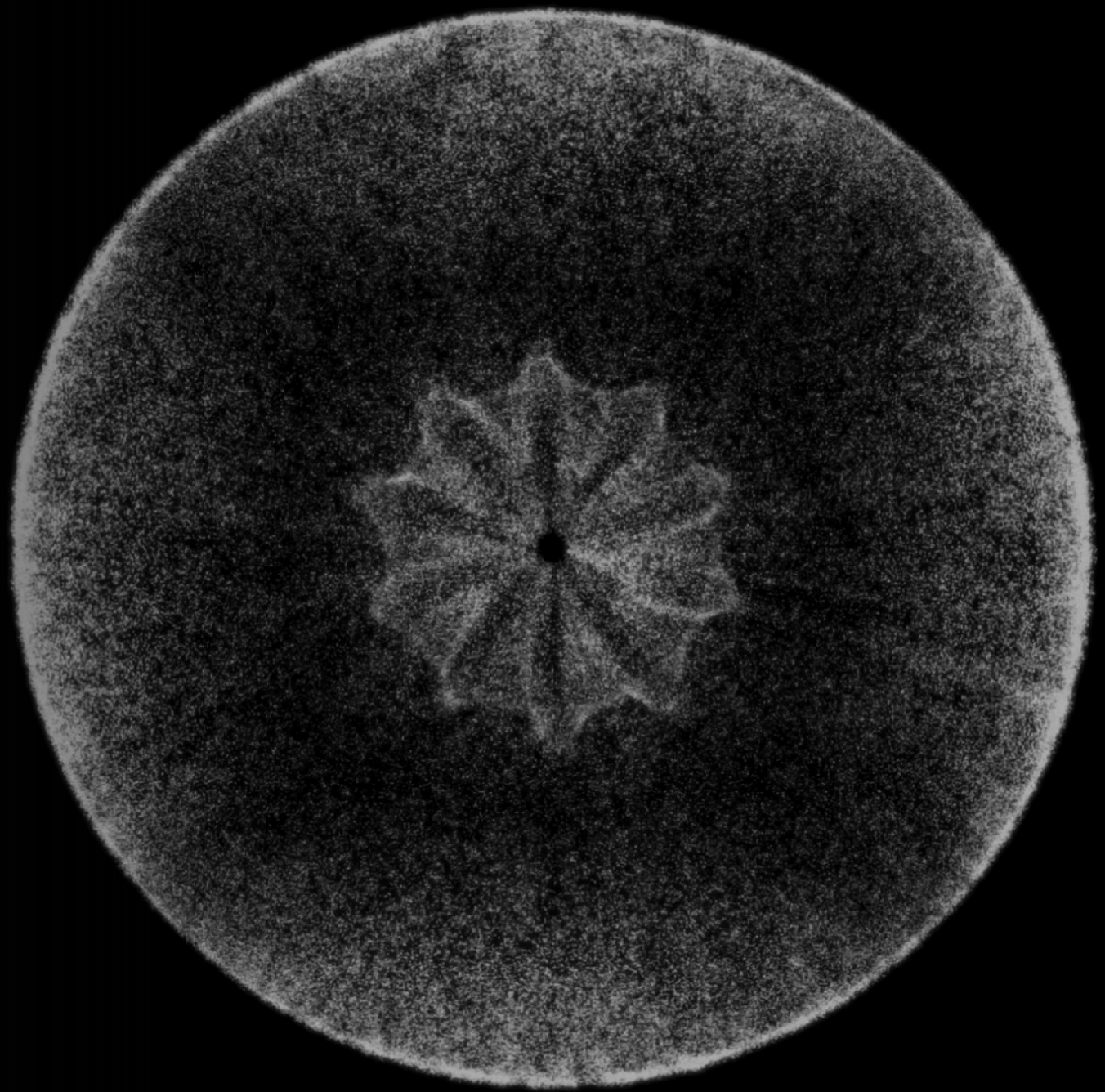
2024 Solitude Internet: filtered seeing and bodily response under the algorithmic landscape. ISAT 2024, 7th International Symposium on Technology Art, TNUA

2021 WHO GOD YOU: a somatic skeleton-recognition game based on machine learning. TAICHI 2021 (demo)

AWARDS

2025 Honorable Mention, CGW 2025 (Paper Shell Echo)

2021 Good Idea Award, Sky Creative Festival (Parkshare proposal)



THANK YOU

Chen Yi-Yen

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Portfolio site & project pages online